WORKED SOLUTION

STEP II 1997, Question 6

Pure • Trigonometric Equations

STEP II 1997, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s2-q6

The key idea

Write everything in terms of $t = \tan\theta$. The triple-angle formula turns the equation into a cubic in t, and once one easy root is spotted the remaining quadratic is exactly the one in the first part, so its roots are handled by a double-angle condition rather than by evaluating $\tan^{-1}(1 \pm \sqrt{2})$. The cotangent equation is the same equation in disguise: the substitution $\theta = \frac{\pi}{2} - \psi$ swaps cot for tan.

Solution

The opening result

Let $t = \tan\phi$ with $t^2 = 2t + 1$. Then $1 - t^2 = -2t$. Also $t \neq 0$, because $t = 0$ does not satisfy $t^2 = 2t + 1$, and $t^2 \neq 1$ (otherwise $2t = 0$). So $\tan 2\phi$ is defined and

$$\tan 2\phi = \frac{2t}{1 - t^2} = \frac{2t}{-2t} = -1,$$

as required.

The converse holds too, and we shall need it. If $\tan 2\phi = -1$ then $2t = -(1 - t^2)$, that is $t^2 = 2t + 1$. So

$$\tan^2\phi = 2\tan\phi + 1 \iff \tan 2\phi = -1.$$

The equation $\tan\theta = 2 + \tan 3\theta$

For a solution, both $\tan\theta$ and $\tan 3\theta$ must be defined. Put $t = \tan\theta$. Then

$$\tan 3\theta = \frac{3t - t^3}{1 - 3t^2},$$

and $1 - 3t^2 \neq 0$ because $\tan 3\theta$ is defined. Hence

$$\tan\theta - \tan 3\theta = \frac{t(1 - 3t^2) - (3t - t^3)}{1 - 3t^2} = \frac{-2t - 2t^3}{1 - 3t^2}.$$

Setting this equal to 2 and multiplying through by $1-3t^2$ (which is non-zero) gives

$$-t-t^3 = 1-3t^2 \quad\Longleftrightarrow\quad t^3-3t^2+t+1=0.$$

The coefficients add up to zero, so $t=1$ is a root, and

$$t^3-3t^2+t+1=(t-1)(t^2-2t-1).$$

Neither $t=1$ nor $t=1\pm\sqrt{2}$ satisfies $3t^2=1$, so every root gives a genuine solution.

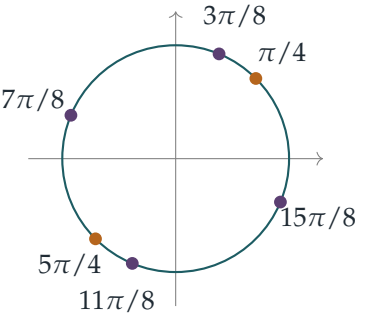
Ask yourself How do we turn $\tan\theta = 1\pm\sqrt{2}$ into rational multiples of π?

We use the first part. The roots of $t^2-2t-1=0$ are exactly the values of $\tan\theta$ with $t^2=2t+1$, and by the equivalence above these are exactly the θ with $\tan 2\theta=-1$.

- $\tan\theta=1$: $\theta=\frac{\pi}{4}+n\pi$. In $(0,2\pi)$ this gives $\theta=\frac{\pi}{4},\ \frac{5\pi}{4}$.
- $\tan 2\theta=-1$: $2\theta=\frac{3\pi}{4}+n\pi$, so $\theta=\frac{3\pi}{8}+\frac{n\pi}{2}$. Here 2θ ranges over $(0,4\pi)$, so $n=0,1,2,3$ and $\theta=\frac{3\pi}{8},\ \frac{7\pi}{8},\ \frac{11\pi}{8},\ \frac{15\pi}{8}$.

(For instance $\tan\frac{3\pi}{8}=1+\sqrt{2}$ and $\tan\frac{7\pi}{8}=-\tan\frac{\pi}{8}=1-\sqrt{2}$.) So the solutions in $0<\theta<2\pi$ are

$$\theta=\frac{\pi}{4},\ \frac{3\pi}{8},\ \frac{7\pi}{8},\ \frac{5\pi}{4},\ \frac{11\pi}{8},\ \frac{15\pi}{8}.$$



The amber points come from $\tan\theta=1$; the plum points, a quarter-turn apart, come from $\tan 2\theta=-1$.

The equation $\cot\theta = 2+\cot 3\theta$

Ask yourself Do we have to redo the whole calculation with cot?

No. Put $\theta=\frac{\pi}{2}-\psi$. Then $\cot\theta=\cot\left(\frac{\pi}{2}-\psi\right)=\tan\psi$. Also

$$\cot 3\theta=\cot\left(\frac{3\pi}{2}-3\psi\right)=\cot\left(\frac{\pi}{2}-3\psi\right)=\tan 3\psi,$$

using the fact that cot has period π. Each side is defined exactly when the corresponding tan is. So the equation becomes

$$\tan\psi = 2 + \tan 3\psi,$$

which is the previous equation. The range $-\frac{3\pi}{2} < \theta < \frac{\pi}{2}$ becomes $0 < \psi < 2\pi$, which is exactly the previous range. So ψ takes the six values found above, and $\theta = \frac{\pi}{2} - \psi$ gives

$$\theta = \frac{\pi}{4},\ \frac{\pi}{8},\ -\frac{3\pi}{8},\ -\frac{3\pi}{4},\ -\frac{7\pi}{8},\ -\frac{11\pi}{8}.$$

Checks

- At $\theta = \frac{\pi}{4}$: $\tan 3\theta = \tan\frac{3\pi}{4} = -1$, and $2 + (-1) = 1 = \tan\frac{\pi}{4}$. At $\theta = \frac{3\pi}{8}$: $\tan\frac{9\pi}{8} = \tan\frac{\pi}{8} = \sqrt{2} - 1$, and $2 + \sqrt{2} - 1 = 1 + \sqrt{2} = \tan\frac{3\pi}{8}$.
- A cubic in t has at most three roots, and tan has period π, so we expect at most $3 \times 2 = 6$ solutions in an interval of length 2π. We found six.
- Numerically, scanning $\tan x - \tan 3x - 2$ on a fine grid over $(0, 2\pi)$ finds sign changes (away from the poles) only at $x = \frac{2\pi}{8}, \frac{3\pi}{8}, \frac{7\pi}{8}, \frac{10\pi}{8}, \frac{11\pi}{8}, \frac{15\pi}{8}$. The same scan for the cotangent equation finds exactly the six values listed.

Common slips

- Stopping at $\tan 2\theta = -1$ with only $n = 0, 1$. Since θ runs over $(0, 2\pi)$, 2θ runs over $(0, 4\pi)$, and that gives four solutions, not two.
- Losing the root $t = 1$ by dividing by something without checking it, or forgetting $\frac{5\pi}{4}$.
- In the last part, solving from scratch and getting the range wrong. The substitution maps the given interval exactly onto $(0, 2\pi)$, and that is no accident.

Beyond STEP

The identity $\tan 3\theta = (3t - t^3)/(1 - 3t^2)$ is the real and imaginary parts of $(1 + \mathrm{i}t)^3$, and in general $\tan n\theta$ is a ratio of polynomials in $\tan\theta$ read off from $(1 + \mathrm{i}t)^n$. The roots of these polynomials are tangents of rational multiples of π, which is why their factorisations, like $t^2 - 2t - 1$ here, are so tidy. Needham's *Visual Complex Analysis* shows how multiple-angle formulae drop out of the geometry of complex multiplication.

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